

FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 – SUB-STRAND 2.1 PROPERTIES OF WAVES: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Today you will write your Final Explanation for the driving question: Why does the matatu's horn change pitch as it passes, and why does the FM radio signal disappear inside the Kabete tunnel, and what do these two mysteries reveal about how ALL waves behave? Answer every section in the space under each prompt. Write in full sentences, use numbers and units where you can, and use evidence from the investigations, simulations and calculations we did in this unit. Draw a labelled diagram wherever a prompt asks for one. Show every step of any calculation. You may use $v = f \lambda$ and the speed of light of 3.0×10^8 m/s.

Part 1: What is a wave? The idea behind both mysteries

A student travelling from Nakuru to Nairobi hears a matatu horn and listens to Classic 105 FM on the bus radio. (a) Explain what a wave is and what it does and does not transfer. (b) Name the wave type (transverse or longitudinal) of the matatu horn sound and of the FM radio signal, and state how the particle or field vibration relates to the direction of travel in each. (c) Explain why it is useful to treat sound and radio as the same kind of thing, even though they seem so different.

Write your answer here:

Part 2: The matatu horn and the Doppler effect

The matatu driver holds the horn at the same setting the whole time, yet the student, in a vehicle travelling in the opposite direction, hears a higher pitch as the matatu approaches and a lower pitch as it moves away. (a) Explain, using what happens to the wave crests, why the detected frequency changes. (b) Say what stays the same and what changes: the horn's own frequency or the frequency detected by the student. (c) Draw and label a diagram showing the wave crests around the moving matatu, marking the direction of the matatu's motion, and mark the positions where the student would hear a higher pitch and where a lower pitch.

Write your answer here:

Part 3: Wavelengths of FM and AM signals

Classic 105 broadcasts at 105.2 MHz. A typical AM station such as KBC Radio broadcasts at 567 kHz. Radio waves travel at 3.0×10^8 m/s. (a) Use $v = f \lambda$ to calculate the wavelength of each signal, showing your working. (b) Compare each wavelength with the width of the Kabete tunnel opening of about 8 m, and state which wave is smaller than the opening and which is larger. (c) A wave travels at constant speed and its frequency is doubled. Describe what happens to its wavelength and explain why.

Write your answer here:

Part 4: Why FM fades and AM survives in the Kabete tunnel

(a) Define diffraction and describe what you found in the single-slit simulation about how the spreading of waves beyond a gap changes with wavelength and with gap width. (b) Use your answers from Part 3 and your diffraction findings to explain why the FM signal fades in the Kabete tunnel while the AM signal remains audible. (c) Our explanation uses a model in which the tunnel opening acts as a gap. Suggest one limitation of this model, and say what other feature of a real tunnel might also affect radio signals. (d) Draw a labelled sketch of the wavefronts of an FM wave and an AM wave meeting the 8 m tunnel opening.

Write your answer here:

Part 5: What the two mysteries reveal about ALL waves

Write a short, connected scientific explanation (6–8 sentences) that answers the driving question. Your explanation should: (a) explain both mysteries using wave properties; (b) show how frequency, wavelength and wave speed link the two; and (c) state one thing you thought at the start of the unit that has changed. Then choose one other wave behaviour you studied in this unit and explain whether it is the cause of the Kabete FM fade, giving a reason.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Understanding of wave concepts	Defines a wave as a disturbance transferring energy without matter and correctly classifies horn (longitudinal) and FM (transverse) with accurate particle motion.	Defines a wave correctly and classifies both waves, with minor gaps in describing vibration direction.	Definition is incomplete or confuses what a wave transfers, or mixes up wave types.
Doppler explanation	Explains compressed and stretched crests, relative motion, and that only the detected frequency changes, with a correct labelled diagram showing both positions.	Explains higher and lower pitch from crest spacing with a reasonable diagram but little emphasis on the unchanged source frequency.	States pitch changes without a mechanism, or suggests the horn itself changes.
Calculations with $v = f \lambda$	Correct substitution, unit conversions and answers (about 2.85 m and 529 m), with the inverse relationship explained.	Correct method with a small arithmetic or unit slip, and mostly correct reasoning about the inverse relationship.	Incorrect method or units, or no link between frequency and wavelength.
Diffraction explanation of the tunnel	Compares wavelength with the 8 m opening for both signals, explains diffraction using simulation findings, acknowledges a limitation of the model, with a clear sketch.	Explains the tunnel using wavelength compared with the opening, with a basic sketch and limited discussion of the model.	Gives a vague or incorrect cause, such as a tunnel simply blocking FM, without comparing wavelength and gap.
Synthesis and scientific communication	Connects both mysteries through shared wave properties, reflects on a change in thinking, and judges another wave behaviour against the tunnel cause using precise vocabulary.	Links both mysteries to wave properties in a clear paragraph with mostly correct terms.	Treats the two mysteries separately, with little use of wave vocabulary or evidence.

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